

Curriculum Vitae

Jorge García Rojas

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Personal Status

Marital Status: Married with a 15 years old daughter

Nationality: Spanish

Date of birth: October 8TH, 1972

Place of birth: La Laguna, S/C de Tenerife (SPAIN)

Present-day Address: San José, 2, 1^oC, 38201. La Laguna. Tenerife. Spain

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Driving License: Yes

Education

- December 18th 2006 – Ph.D. in astrophysics: *Chemical abundances in Galactic H II regions and temperature fluctuations*. University of La Laguna (ULL), Tenerife, Spain.
Maximum qualification: *Apto Cum Laude*.
- July 1998 – Master's Degree: *Chemical abundances on the Galactic H II region M8*. ULL, Tenerife, Spain.
- December 1996 – Post Graduate Certificate in Education. Faculty of Education, University of Salamanca, Spain.
- March 1996 – BSc on Physics/Astrophysics. ULL, Tenerife, Spain.

Prizes

- University prize for best thesis in Experimental Sciences (3 prizes per year). ULL, Tenerife, Spain. 2007

Working Experience

- June 2026 – Now: Postdoc Plan Nacional project
- December 2024 – June 2026: Postdoc related to scientific exploitation of MAAT@GTC instrument at the IAC, Spain
- November 2016 – December 2024: Advanced Postdoc “Severo Ochoa” position at the IAC, Spain
- February 2014 – December 2015: Postdoc “Severo Ochoa” at the IAC, Spain.
- June 2009 – February 2014: Support Astronomer of the IAC observatories. Spain.
- October 2007 – May 2009: Postdoctoral fellowship in the Instituto de Astronomía. Universidad Nacional Autónoma de México (IA-UNAM). México City. México.
- June 2007 – July 2007: Visiting fellowship in the Instituto de Astronomía, Óptica y Electrónica (INAOE) Tonanzintla, Puebla. México.
- May 2007 – June 2007: Visiting fellowship in the IA-UNAM. México City. México.
- February 2007 – April 2007: Visiting fellowship in the IAC, Spain.
- October 2002 – September 2006: FPI fellowship of the Ministerio de Ciencia y Tecnología at the IAC, Spain. PhD in Astrophysics, under the supervision of Dr. César Esteban López with the Thesis entitled: “Chemical abundances in H II regions and temperature fluctuations.”

- November 2001 – March 2002: Monitor in Astronomical Events at the *Sciences and Cosmos Museum of Tenerife*.
- October 1999 – August 2001: Sciences teacher at the *Tinajo Secondary School* in Tinajo (Lanzarote, Spain)
- October 1998 – August 1999: Sciences teacher at the *Mayex school* in La Laguna (Tenerife, Spain)
- June–July 1998 and March–September 2002: Fellowship in the “Arturo Duperier Observatory” at Arrecife (Lanzarote, Spain). Teaching astronomy to kids and tourists.
- May and September 1998: Scholarship at the Computer Room at the Faculty of Physics of the ULL. Computing support to students.

Observing experience

I have a huge observing experience using mid-high resolution spectroscopic facilities. I have granted observing time as P.I. or co-I in several of the largest optical telescopes:

- ALFOSC (spectroscopy and image modes) and FIES spectrograph at the 2.5m NOT, Roque de los Muchachos Observatory, La Palma, Spain.
- PMAS IFU spectrograph at the 3.6m at Calar Alto Observatory, Almeria, Spain
- SARG echelle spectrograph at the 3.6 Galileo telescope, Roque de los Muchachos Observatory, La Palma, Spain.
- ISIS spectrograph and OASIS+NAOMI integral field spectrograph with adaptive optics (AO) at the 4.2m WHT, Roque de los Muchachos Observatory, La Palma, Spain.
- MIKE spectrograph at the 6.5m Magellan telescope. Las Campanas Observatory, Chile.
- HDS spectrograph at the 8m Subaru telescope, Mauna Kea Observatory, Hawaii, USA.
- UVES, FORS2, FLAMES, and MUSE spectrographs at the 8m VLT. Cerro Paranal Observatory, Chile.
- HIRES spectrograph at the 10m Keck II, Mauna Kea Observatory, Hawaii, USA.
- OSIRIS, EMIR and MEGARA at the 10.4m GTC, Roque de los Muchachos Observatory, La Palma, Spain.

Additionally, I have huge experience with several facilities in the Teide and Roque de los Muchachos observatories during my support astronomer years. I have more than 200 nights experience using different instruments at IAC80 and TCS telescopes at the Teide observatory, and at INT, WHT, TNG and NOT at Roque de Los Muchachos observatory.

Visiting researcher

- 15 Feb 2023 – 1 Mar 2023 – National Observatory of Athens (Athens, Greece), working with Dr. Stavros Akras.
- 31 Oct 2019 – 8 Nov 2019 – Instituto de Astronomía-UNAM (Ensenada, Baja California, Mexico), working with Dr. Christophe Morisset.
- 20 Oct 2019 – 30 Oct 2019 – Instituto de Astronomía-UNAM (Mexico D.F., Mexico), working with Dr. Gloria Delgado-Inglada and Dr. Grazyna Stasinska.
- 16 Apr 2018 – 20 Apr 2018 – Observatory of Paris-Meudon (Paris, France), working with Dr. Grazyna Stasinska and Dr. Gloria Delgado-Inglada.
- 12 Jan 2018 – 19 Jan 2018 – Observatory of Paris-Meudon (Paris, France), working with Dr. Grazyna Stasinska and Dr. Gloria Delgado-Inglada.
- 15 Oct 2017 – 20 Oct 2017 – Instituto de Astronomía-UNAM (Mexico D.F., Mexico), working with Dr. Christophe Morisset and Dr. Gloria Delgado-Inglada.
- 18 May 2015 – 22 May 2015 – Instituto de Astronomía, Pontificia Universidad Católica (Santiago de Chile, Chile), working with Dr. Adal Mesa-Delgado.

- 9 Mar 2015 – 28 Mar 2015 – Instituto de Astronomía-UNAM (Mexico D.F., Mexico), working with Dr. Miriam Peña, Dr. Christophe Morisset and Dr. Gloria Delgado-Inglada.
- 6 May 2013 – 24 May 2013 – Instituto de Astronomía-UNAM (Mexico D.F., Mexico), working with Dr. Miriam Peña, Dr. Christophe Morisset and Dr. Gloria Delgado-Inglada.
- 16 May 2011 – 27 May 2011 – Instituto Nacional de Astrofísica, Óptica y Electrónica (Tonanzintla, Puebla. Mexico) working with Dr. Mónica Rodríguez.
- 18 Apr 2011 – 15 May 2011 – Instituto de Astronomía-UNAM (Mexico D.F., Mexico) working with Dr. Miriam Peña and Prof. Manuel Peimbert.
- 1 Nov 2010 – 30 Nov 2010 – Instituto de Astronomía-UNAM (Mexico D.F., Mexico) working with Dr. Miriam Peña and Dr. Christophe Morisset.
- Dec 2008 – Jan 2009 – Instituto de Astrofísica de Canarias, working with Dr. Sergio Simón-Díaz.
- Dec 2007 – Jan 2008 – Instituto de Astrofísica de Canarias, working with Dr. César Esteban.
- 20 Jun 2007 – 20 Jul 2007 – Instituto Nacional de Astrofísica, Óptica y Electrónica (Tonanzintla, Puebla. Mexico) working with Dr. Mónica Rodríguez.
- 20 May 2007 – 20 Jun 2007 – Instituto de Astronomía-UNAM (Mexico D.F., Mexico) working with Prof. Manuel Peimbert, Dr. Miriam Peña and Dr. Leticia Carigi.
- 22 Oct 2005 – 6 Nov 2005 – Instituto Nacional de Astrofísica, Óptica y Electrónica (Tonanzintla, Puebla. Mexico) working with Dr. Mónica Rodríguez.
- 14 Oct 2005 – 21 Oct 2005 – Instituto de Astronomía-UNAM (Mexico D.F., Mexico) working with Prof. Manuel Peimbert, Dr. Antonio Peimbert and Dr. Leticia Carigi.
- 28 Feb 2004 – 5 Mar 2004 – Instituto Nacional de Astrofísica, Óptica y Electrónica (Tonanzintla, Puebla. Mexico) working with Dr. Mónica Rodríguez.
- 20 Feb 2004 – 27 Feb 2004 – Instituto de Astronomía-UNAM (Mexico D.F., Mexico) working with Prof. Manuel Peimbert, Dr. Antonio Peimbert and Dr. Leticia Carigi.

Language Skills

- Spanish (Mother tongue)
- Spoken and written English. Advanced degree of the Spanish Official Languages School.

Research interests

- Chemical abundances in ionized nebulae / Abundance discrepancies in ionized nebulae.
- 2D spectroscopy of photoionized nebulae.
- *s*-processes in Asymptotic Giant Branch Stars (AGBs)
- Ionization structure of H II regions.
- Galactic chemical evolution
- High-resolution spectroscopy.
- Photoionization modelling.

Invited colloquia and seminars

- 29 May 2026 – “*The DESIRED project: Physical conditions, chemical abundances, and metallicities in photoionized nebulae*”, seminar at the “Stellar and Interstellar Physics “ Severo Ochoa Research Line monthly meeting, IAC, Tenerife, Spain
- 10 Apr 2024 – “*Una historia sobre la evolución química del Universo*” invited talk at the “XIX Congreso de Estudiantes de la Sección de Química de la ULL”. Facultad de Ciencias, sección Química, ULL, Tenerife, Spain
- 22 Feb 2023 – “*The abundance discrepancy problem in photoionized nebulae through the eyes of a MUSE*”, seminar at National Observatory of Athens, Athens, Greece.

- 13 Oct 2022 – “*The abundance discrepancy problem in photoionized nebulae through the eyes of a MUSE*”, online seminar at the Instituto de Radioastronomía y Astrofísica (IRyA), Morelia, Mexico.
- 27 Jun 2022 – “*The abundance discrepancy problem in photoionized nebulae through the eyes of a MUSE*”, online seminar at the Instituto de Astronomía de la UNAM. Sede Ensenada, Baja California, Mexico.
- 8 Jun 2022 – “*The abundance discrepancy problem in photoionized nebulae through the eyes of large telescopes*”, “GTC Academy seminar”, online seminar for the Chinese Academy of Sciences.
- 27 Jun 2019 – “*The abundance discrepancy phenomenon in planetary nebulae: recent results*”, “Thirty minutes talk” at ESO Headquarters in Santiago de Chile, Chile
- 24 May 2017 – “*The large abundance discrepancy phenomenon in planetary nebulae*”, seminar at the Department for Theoretical Physics at the Universidad Autónoma de Madrid, Spain
- 27 Apr 2017 – “*The high abundance discrepancy phenomenon in planetary nebulae*”, seminar at Instituto de Astrofísica de Andalucía, Granada, Spain
- 28 Apr 2017 – “*Abundancias de elementos químicos en nebulosas fotoionizadas*”, talk at Master FISYMAT of the University of Granada, Spain
- 1 Jul 2016 – “*Exploring the link between high ADF planetary nebulae and close binary stars*”, “Thirty minutes talk” at ESO Headquarters in Santiago de Chile, Chile
- 12 Nov 2015 – “*Clues on the origin of the abundance discrepancy factor in planetary nebulae*”, general talk at Osservatorio Astrofisico di Arcetri, Firenze, Italy
- 22 May 2015 – “*Faint emission lines in very deep UVES spectra of photoionized nebulae*”, “Thirty minutes talk” at ESO Headquarters in Santiago de Chile, Chile
- 20 May 2015 – “*Binarity and the abundance discrepancy in planetary nebulae*”, general colloquium at IA-PUC, Santiago de Chile, Chile
- 25 Mar 2015 – “*Binarity and the abundance discrepancy in planetary nebulae*”, IA-UNAM colloquium at IA-UNAM, Mexico D.F., Mexico.
- 18 Mar 2015 – “*Binarity and the abundance discrepancy in planetary nebulae*”, institutional seminar at INAOE, Puebla, Mexico.
- 20 Sep 2013 – “*Estado de la instrumentación de los telescopios del ORM/OT para los próximos semestres*”, seminar given by the IAC Support Astronomers Group, and GTC support astronomers at IAC, Tenerife, Spain.
- 16 Mar 2012 – “*Status update and instrumentation prospects for GTC and other ORM/OT telescopes for the coming semesters*”, seminar given by the IAC Support Astronomers Group and GTC operations manager at IAC, Tenerife, Spain.
- 19 Jul 2011 – “*M43, the little sister of the Orion nebula*”, “Breaking News” seminar at IAC, Tenerife, Spain. Given in collaboration with Sergio Simón-Díaz.
- 18 May 2011 – “*M43, the little sister of the Orion nebula*”, IA-UNAM colloquium at IA-UNAM, Mexico D.F., Mexico.
- 9 May 2011 – “*M43, the little sister of the Orion nebula*”, institutional seminar at INAOE, Puebla, Mexico.
- 3 October 2008 – “*Unveiling the physical and chemical properties of the object HH202 in the Orion nebula through 2D and echelle spectrophotometry*”, institutional seminar at INAOE, Puebla, Mexico
- 22 Sep 2008 – “*2D and echelle spectroscopy of the objet HH202 in the Orion nebula*”, DEMI colloquia at IA-UNAM, Mexico D. F., Mexico.
- 13 Jul 2007 – “*Chemical abundances in HII regions and temperature fluctuations*”, institutional seminar at INAOE, Puebla, Mexico.
- 11 Jul 2007 – “*Determination of chemical abundances in HII*”, informal seminar at “Coffee and astronomy” students meeting at INAOE, Puebla, Mexico.
- 15 Jun 2007 – “*Chemical abundances in HII regions and temperature fluctuations*”, at CRyA, Morelia, Mexico.

- 13 Jun 2007 – “*Chemical abundances in HII regions and temperature fluctuations*”, at IA-UNAM, Mexico D.F., Mexico.
- 9 Feb 2006 – “*Chemical abundances in Galactic HII regions*”, general seminar at IAC, Tenerife, Spain.
- 21 Oct 2005 – “*Physical and Chemical properties of the Orion Nebula. A correlation with some morphological structures*”, IA-UNAM colloquium at IA-UNAM, Mexico D.F., Mexico.

Teaching

- 2014/15, 2015/16, 2017/18, 2018/19, 2021/22, 2022/23, 2023/24, 2024/25, 2025/26 – Assistant teacher of the subject “Scientific Computation I” 1st course of the Physics Grade at the ULL.
- “Chemical abundances determinations in photoionized regions” Two lectures (three hours) of a master level course at the IA-UNAM, Mexico. November 2008.
- “Chemical abundances determinations in photoionized nebulae”. Talk for the master’s level course “Complementary research activities” at the ULL. Each school year since 2012-13.
- Three years of experience in secondary schools as a full-time sciences (mathematics, physics, chemistry and technology) teacher.

Students and postdocs supervision

Postdocs

- David Jones
IAC / Centro de Astrofísica de La Palma (CALP)
December 2018 – December 2020
Current professional status: Permanent position at the IAC.
- Verónica Gómez-Llanos
IAC
Mayo 2022 – Mayo 2023
Current professional status: Left academy

PhD students

- Maialen Orte García
Co-supervised with Dr. César Esteban (IAC, Spain)
Internal and global properties of local star-forming nebulae (HII regions)
ULL/IAC
Starting date: 1 April 2024.
- Jose Eduardo Méndez Delgado
Co-supervised with Dr. César Esteban (IAC, Spain)
Ionized gas flows in the Orion Nebula: properties and environmental dependences
ULL/IAC
Date of defense: 23 September 2022.
Maximum qualification with honors (Sobresaliente Cum Laude)
Current professional status: Tenure-track position at IA-UNAM, México.
- Gisela Domínguez Guzmán
Co-supervised with Dr. Mónica Rodríguez (INAOE, Mexico) and Dr. César Esteban (IAC, Spain)
Iron depletion onto dust in low metallicity photoionized regions
INAOE, Puebla, México.

Starting date: 01/09/2015.

Current status: Temporarily paused due to extenuating family health circumstances.

- Simone Madonna
Supervisor with Dr. Nicholas C. Sterling (co-supervisor, U. West Georgia, USA)
Study of the chemical abundances of s-elements in planetary nebulae.
ULL/IAC
Date of defense: 31 January 2019.
Maximum qualification with honours (Sobresaliente Cum Laude)
Current professional status: Left academia to pursue opportunities in education and science outreach. Lecturer at the International University of La Rioja. Developing a PhD in Philosophy of Science
- Laura Toribio San Cipriano
Co-supervised with Dr. César Esteban (IAC, Spain)
Physical conditions and chemical abundances in photoionized nebulae.
ULL/IAC
Date of defense: 21 July 2017.
Maximum qualification with honours (Sobresaliente Cum Laude)
Current professional status: Permanent position at CIEMAT, Spain

Master students

- Elena Reyes Rodríguez
Supervisor with Dr. J. E. Méndez-Delgado (co-supervisor)
Investigating the excitation mechanism of the C II $\lambda 6578$ line in Planetary Nebulae
ULL
March 2023
Qualification: 9.0/10.0
Current professional status: Contract as an engineer in optical communications at IAC-Tec (Tenerife, Spain)
- Joanna Sakowska
Supervisor with Dr. D. Aníbal García-Hernández (co-supervisor)
Constraining the Stellar Progenitors of Double-Dust Chemistry Planetary Nebulae
University of Surrey, UK/IAC
February 2020
Qualification: 78% (First-class)
Current professional status: Postdoc at IAA, Granada, Spain
- Juan Bautista Climent Oliver
Supervisor with Dr. C. Esteban (co-supervisor)
Spatial variations of the abundance discrepancy in the planetary nebula M 1-42
ULL
June 2016
Qualification: 9.8/10.0
Current professional status: Assistant professor at the University of Valencia, Spain
- David Tejera Flores
Co-supervisor with Dr. C. Esteban
Chemical abundances in H II regions of the Magellanic Clouds
ULL
July 2016
Qualification: 5.0/10.0
Current professional status: Left academia to pursue opportunities in education

- Gisela Domínguez Guzmán
Co-supervisor with Dr. Mónica Rodríguez (INAOE, Mex) and Dr. César Esteban (IAC)
Iron depletion onto dust in HII regions and planetary nebulae of the Magellanic Clouds
INAOE, Puebla, México
August 2015
Qualification: Qualified
Current professional status: PhD in progress. Temporarily paused due to extenuating family health circumstances.
- Víctor Pérez-Mesa
Co-supervisor with Dr. César Esteban (IAC, Spain)
The Chlorine Abundance gradient in the Milky Way
ULL
March 2015
Qualification: 9.0/10.0
Current professional status: PhD completed at ULL/IAC. Left academia to pursue opportunities in industry.
- Simone Madonna
Co-supervisor with Dr. Valentina Luridiana
Study of the chemical abundances of s-elements in planetary nebulae
ULL
June 2014
Qualification: 9.5/10.0
Current professional status: PhD completed at ULL/IAC. Left academia to pursue opportunities in education and science outreach. Lecturer at the International University of La Rioja. Developing a PhD in Philosophy of Science

Undergraduate thesis students

- Kevin Nicolás Álvarez Báez
Supervisor with MSc Maialen Orte García (co-supervisor)
The chlorine abundance in planetary nebulae
ULL
June 2026
Qualification: 8.3/10.0

IAC Summer students

- Carlos José Díaz
Co-supervisor with Dr. Cristina Zurita (IAC, Spain)
Development and automation of the data analysis for an Exposure Time Calculator in CAMELOT and CAIN
July-September 2013
- Adal Mesa-Delgado
Co-supervisor with Dr. César Esteban (IAC, Spain)
Physical and chemical properties of the ionized gas in the Orion Nebula and their correlation with different morphological structures
July-September 2005.

Workshops, conferences and meetings attended

- October 2026 – “*Tercer Congreso Nacional de Estrellas Evolucionadas*”, Mexico
Invited talk: “*Chemical abundance studies in planetary nebulae from optical integral field spectroscopy*”
- July 2026 – “*Guillermo Haro Workshop 2026. Chemical abundances, ISM properties, and galaxy evolution across cosmic time*” Tonanzintla, Puebla, Mexico
Contributed talk: “*Integral field spectroscopy of planetary nebulae. From chemical abundances to low ionization structures*”
- September 2025 – “*Asymmetrical Post-main-sequence Nebulae 9: The Art of Stellar Wind Sculpting*” Granada, Spain.
Contributed talk: “*2D spectroscopy of Galactic planetary nebulae with high abundance discrepancies: Current status and future prospects*”
- November 2024 – “*A decade of discoveries with MUSE and beyond*” Garching, Germany.
Contributed talk: “*A MUSE view of planetary nebulae with large abundance discrepancies*”
- July 2024 – “*XVI Meeting of the Spanish Astronomical Society*” Granada, Spain.
Contributed talk: “*WEAVE Science verification LIFU data of Galactic planetary nebulae*”
- May 2024 – “*IV Workshop on Chemical abundances in gaseous nebulae*” Sao Jose dos Campos, Sao Paulo, Brazil.
Contributed talk: “*MUSE observations of high abundance discrepancy planetary nebulae: when abundance discrepancies turn to be abundance contrasts between two different gas phases*”
- September 2023 – “*IAU Symposium 384: Planetary Nebulae: a Universal Toolbox in the Era of Precision Astrophysics*”
Invited talk: “*The abundance discrepancy in ionized nebulae: which are the correct abundances?*” RENOUNCED
- September 2022 – “*XV Meeting of the Spanish Astronomical Society*” La Laguna, Tenerife, Spain.
Contributed talk: “*High-ADF planetary nebulae through the eyes of a MUSE*”
- May 2022 – “*VII GTC Science Meeting*” Virtual meeting
Contributed talk: “*The most luminous planetary nebulae in M31 unravelled by deep GTC-OSIRIS spectroscopy*”
- May 2021 – “*III Workshop on Chemical abundances in gaseous nebulae*” Virtual meeting
Contributed talk: “*Neutron capture elements in planetary nebulae*”
- March 2021 – “*MAAT Data System Workshop*” Virtual meeting
Contributed talk: “*Galaxies & Nebulae Data System Specs*”
- March 2019 – “*II Workshop on Chemical abundances in gaseous nebulae*” Sao Jose dos Campos, Sao Paulo, Brazil.
Invited talk: “*Latest advances in the abundance discrepancy problem in photoionized nebulae*”
- August 2018 – “*Radial metallicity gradients in star forming galaxies. Focus meeting 7 at the IAU General Assembly*” Viena, Austria.
Invited talk: “*Radial metallicity gradients from Galactic nebular probes*”
- October 2017 – “*The Cosmic Feast of the Elements. A conference to celebrate the work of Grazyna Stasinska*” Puebla, Mexico.
Contributed talk: “*How binarity affect the abundance discrepancy in planetary nebulae*”
- September 2017 – *Workshop* “*Proposing and processing MEGARA observations with GTC*”, Madrid, Spain.
Talk: “*MEGARA: a workhorse instrument for the physics of planetary nebulae*”
- November 2016 – *Workshop* “*Chemical abundances in gaseous nebulae*” Campos do Jordão, Sao Paulo, Brazil.
Invited talk: “*Chemical abundances in planetary nebulae from faint emission lines*”
- October 2016 – “*IAU Symposium 323. Planetary nebulae: Multi-wavelength probes of stellar and Galactic evolution*” to be held in Beijing, China.

- Contributed talk: *“Imaging the elusive H-poor gas in high ADF planetary nebulae”*
- August 2016 – *“Cloudy: Emission lines in astrophysics, from gaseous nebulae to quasars. A symposium to honor Gary Ferland”* Mexico D. F., Mexico.
Contributed talk: *“Exploring the link between high ADF planetary nebulae and close binary central stars”*
 - February 2016 – *Workshop “1st RIA-MEGARA Open day”*, Madrid, Spain.
 - December 2015 – *“Vth Science with GTC”*. Puebla, Mexico.
Contributed talk: *“Imaging the elusive H-poor gas in high ADF planetary nebulae with GTC”*
 - September 2014 – *“XI Scientific meeting of the Spanish Astronomical Society”*, Teruel, Spain.
Contributed talk: *“Ultra-deep high-resolution spectra of planetary nebulae: s-process enrichment in NGC 3918”*
Poster: *“Do stellar and nebular abundances in the Cocoon nebula agree?”*
 - November 2013 – *“Asymmetrical Planetary Nebulae VI”*, Playa del Carmen, Mexico.
Poster: *“Deep high-resolution spectroscopy of a sample of planetary nebulae ionized by [WC] central stars”*
 - July 2012 – *“2nd NCAC Symposium: The Orion Nebula: A Laboratory for the Study of Star Formation and Gaseous Nebulae”*, Warsaw, Poland.
Contributed talk: *“M43, the little sister of the Orion nebula”*
 - May 2012 – *“Mapping the Oxygen in the Universe”*, Puerto de la Cruz, Tenerife, Spain.
Contributed talk: *“Oxygen abundances from optical O I and O II recombination lines in planetary nebulae with [WC] central stars”*
 - July 2011 – *“IAU Symposium 283. Planetary Nebulae: An Eye to the Future”*, Puerto de la Cruz, Tenerife, Spain.
Poster: *“Abundances and ADFs in PNe with [WC] central stars”*
 - November 2010 – *“XIII Latin-American Regional IAU Meeting”*, Morelia, Mexico. Contributed talk: *“Faint emission lines in PNe with [WC] nucleus”*
 - October 2010 – *“Uncertainties in atomic data and how they propagate in chemical abundances”*, La Laguna, Tenerife, Spain.
 - June 2008 – *“The Cosmic Odyssey of the elements”*, Aegina, Greece.
Invited talk: *“Observational results on the abundance discrepancy problem in HII regions”*
 - April 2008 – *“XXII Congreso Nacional de Astronomía”*, Universidad Iberoamericana, Mexico D.F. Mexico.
Contributed talk: *“Small scale variations of the physical conditions and the abundance discrepancy in the Orion nebula”*
 - April 2007 – *Workshop “Deep Spectroscopy and Modelling of Gaseous Nebulae”*, Xiang Shan, Beijing, China.
Contributed talk: *“H II region abundances from recombination lines. The abundance discrepancy problem”*
 - January 2007 – *Workshop “Metallicity calibrations for gaseous nebulae”*, Institute for Astronomy at Manoa, Honolulu, Hawaii, USA.
Invited talk: *“H II region metallicities from recombination lines”*
 - September 2006 – *“VII SEA Scientific Meeting”*, Barcelona, Spain.
Contributed talk: *“Abundancias Químicas en regiones H II galácticas”*
 - June 2006 – *“The Metal Rich Universe”*, La Palma, Spain.
 - September 2004 – *“The many scales in the Universe. Joint European and National Astronomy Meeting”*, Granada, Spain.
Poster: *“Tailored models of the H II region S 311”*
 - February 2004 – *“II International GTC Workshop: Science with GTC 1st-light Instruments and the LMT”*, Mexico DF, Mexico.
Poster: *“Faint C and O recombination lines in H II regions and large telescopes: The case of S311”*

Organization of international conferences

- 23 – 27 October 2017 – Co-chair of the Scientific Organizing Committee of the international conference “*The cosmic feast of the elements: a conference to honor Grazyna Stasinska*”, Puebla, Mexico.
- 22 –26 June 2015 – “*European Week of Astronomy and Space Sciences (EWASS) 2015*” Co-chair of the Scientific Organizing Committee in the special session 18 “*Chemical abundances and gradients in spatially resolved late-type galaxies in the local Universe*” La Laguna, Tenerife, Spain.
- 14 – 18 May 2012– Chair of the Local Organizing Committee of the international conference “*Mapping the Oxygen in the Universe*” Puerto de la Cruz, Tenerife, Spain
- 25 – 27 October 2010 – “*Uncertainties in atomic data and how they propagate in chemical abundances*”, Tenerife, Spain. Local Organizing Committee member (LOC)

Principal Investigator on research and technical projects

Project. “*Dissecting the binary cores of planetary nebulae: a novel view of chemical abundances and gas kinematics*”. (AYA2017-83383-P)

Research Center. Instituto de Astrofísica de Canarias

Duration. 01/01/2018- 31/12/2020

Project Manager. Dr. **Jorge García Rojas** (PI) and Dr. Pablo Rodríguez Gil (co-PI)

Total Funds: 145200 €

Project. *Espectroscopía 2D en nebulosas planetarias con alta discrepancia de abundancias*”. (ProID2021010074)

Research Center. Instituto de Astrofísica de Canarias

Duration. 01/05/2021- 30/04/2023

Project Manager. Dr. **Jorge García Rojas** (PI)

Total Funds: 69050 €

I additionally actively participate or have participated as co-I in other 23 research projects funded in competitive calls. 9 of them at the IAC, 12 at different Mexican institutions (UNAM, INAOE), 1 at Greece (IAASARS) and 1 at USA (U. West Georgia).

National co-coordinator of ERASMUS+ Projects

Project. “*European Collaborating Astronomer ProjectS: Espana-Czechia-Slovakia*”. (2020-1-CZ01-KA203-078200)

Research Center. Institute of Astronomy. Czech Academy of Sciences, Ondrejov, Prgue, Czech Republic

Duration. 01/09/2020- 31/08/2023

Project Manager. Dr. Petr Kabath

Role on Project: Co-coordinator at the IAC

Total Funds: 318085 €

Project. “*Per aspera ad astra simul (Through difficulties to the stars together)*”. (2017-1-CZ01-KA203-035562)

Research Center. Institute of Astronomy. Czech Academy of Sciences, Ondrejov, Prgue, Czech Republic

Duration. 01/10/2017- 30/11/2020

Project Manager. Dr. Petr Kabath

Role on Project: Co-coordinator at the IAC

Total Funds: 288164 €

Outreach talks and activities.

- I am active in giving outreach talks for different levels: preschoolers, primary and secondary students, university students, and the general public. In total, I have given more than 15 outreach talks.
- I have been involved in the design and writing of 9 press releases at the IAC.
- I have mentored last-year high school students interested in studying physics or astrophysics for the event “Canarias Masterclass” for three years since 2022, mentoring a total of 5 students.
- I was involved in two large-scale astronomy outreach events in México: in February 2008, I attended the “*Eclipse en el Zócalo*” event in México D.F., giving talks to small groups about constellations and planets, and in November 2009, I attended the “*Noche de las estrellas*” event in Monte Albán, Oaxaca, México, giving the talk “*The life of the stars*”.

Membership of professional societies

- Senior member of the Spanish Astronomical Society
- Senior member of the European Astronomical Society. ID #2745
- Member of the International Astronomical Union. Member ion divisions C (Education, Outreach and Heritage), G (Stars and Stellar Physics), and H (Interstellar Matter and Local Universe) and commission H3 (planetary nebulae). I am right now elected member of the Commission Organizing Committee of the H3 Planetary Nebulae subdivision.

Other merits

- I regularly referee papers for *The Astrophysical Journal*, *The Astronomical Journal*, *Astronomy & Astrophysics*, and *Monthly Notices of the Royal Astronomical Society*.
- I have been an external reviewer of research proposals for the Scientific and Technological Research Council of Chile (CONICYT), for the “Programa de Apoyo a Proyectos de Investigación e Innovación Tecnológica” (PAPIIT) of the Universidad Nacional Autónoma de México (UNAM), and for the Spanish National Research Agency (ANECA, Spain).
- I have been an external referee for several proposals for the OPTical Infrared Co-Ordination Network (OPTICON)
- I have been a member of the Time Allocation Committee (TAC) of Canary Island observatories during 4 semesters: 2019B, 2020A, 2020B and 2021A.
- I have been a reviewer for an OPC/Expert Panel in Category C for ESO proposals in periods 116 and 117.
- I have been a member of the GTC Users Committee (GUC) between July 2018 and February 2020.
- I have huge experience with non-professional telescopes and astronomy outreach.
- I regularly write outreach articles for the “*Gaveta de Astrofísica*” section of the local newspaper “*El Día*” (see outreach publication list)

List of Publications

Papers in refereed journals

1. Arellano-Córdova, K. Z.; Flury, S. R.; Cullen, F.; Esteban, C.; Vilchez, J. M.; **García-Rojas, J.**; et al. (+14 authors)
Unveiling extreme $T_e[O III] I$: The evolution of the electron temperature and metallicity across redshifts
2026, MNRAS, submitted
2. Mousley, A.; Jones, D.; Dorsch, M.; Manick, R.; Wesson, R.; Reindl, N.; Escorza, A.; Sowicka, P.; Boffin, H. M. J.; Chornay, N.; Csukai, A.; **García-Rojas, J.**
The long-period central star of the planetary nebula MPAJ1508–6455
2026, A&A, submitted
3. Méndez-Delgado, J. E. et al. (+31 authors, including **García-Rojas, J.**)
[*There is no single density: star-forming regions and galaxies hold more dense ionized gas than long assumed*](#)
2026, Nature, submitted, arXiv:2607.07973
4. Rosales-Ortega, F. F.; Méndez-Delgado, J. E.; Guerrero-González, J. U.; Esteban, C.; **García-Rojas, J.**; et al. (+14 authors)
[*The DESIRED strong-line calibrations: I. New empirical metallicity relations for the local and high-redshift Universe*](#)
2026, submitted, arXiv:2604.16273
5. Konstantinou, L.; Akas, D.; Bouvis, K.; Boumis, P.; **García-Rojas, J.**; Gonçalves, D. R.; Gourgouliatos, K. N.
X-Shooter view of planetary nebulae microstructures I. The case of NGC 7009 knots
2026, A&A, in press, arXiv:2609.
6. Sterling, N. C.; Stephenson, M. G.; Dinerstein, H. L.; Yagüe López, A.; Kaplan, K. F.; Beaumont, J.; Lugaro, M.; Karakas, A. I.; Battino, U.; **García-Rojas, J.**; Erben, S. M. H.
[*The s-process at Subsolar Metallicity: Insights from High-Resolution Infrared Spectroscopy of Magellanic Cloud Planetary Nebulae*](#)
2026, ApJ, **286**, 47
7. Merc, J. et al. (+25 authors, including **García-Rojas, J.**)
[*K 1-6 is a photoionized ISM nebula shaped by a fast-moving hot white dwarf in a triple system*](#)
2026, A&A, **713**, A283
8. **García-Rojas, J.**; Reyes-Rodríguez, E.; Méndez-Delgado, J. E.; Morisset, C.; et al. (+6 authors)
[*Investigating the electron temperature of \[Ar IV\] in planetary nebulae using the DESIRED database*](#)
2026, A&A, **713**, A97
9. Amayo, A.; Carigi, L.; Méndez-Delgado, J. E.; **García-Rojas, J.**; Reyes-Rodríguez, E.
[*A comparative study of O, Ne, Cl, and Ar in H II regions and PNe of the galactic disk: Temporal evolution of radial gradients?*](#)
2026, MNRAS, **550**, 2, stag1196
10. Orte-García, M.; Esteban, C.; **García-Rojas, J.**; Méndez-Delgado, J. E. et al. (+ 6 authors)
[*The DESIRED electron temperature relations in star-forming regions of the local Universe*](#)
2026, A&A, **711**, A165
11. Arellano-Córdova, K. Z.; Méndez-Delgado, J. E.; Flury, S. R.; Esteban, C.; Kreckel, K.; **García-Rojas, J.**; et al. (+ 7 authors)
[*A self-consistent direct method for chemical abundances in high-z galaxies with JWST*](#)
2026, MNRAS, **547**, 4, stag380
12. Jones, D.; Corradi, R. L. M.; García-Pérez, G. A.; Morisset, C.; **García-Rojas, J.**; et al. (+13 authors)
[*Appraising the Necklace: A post-common-envelope carbon dwarf inside an apparently carbon-poor planetary nebula*](#)
2026, A&A, **707**, A169
13. Martínez-Hernández, I. R.; Méndez-Delgado, J. E.; Esteban, C.; **García-Rojas, J.** et al. (+16 authors)
[*The DESIRED temperature–metallicity relations in star-forming regions: probing the Galactic radial and azimuthal metallicity distributions*](#)
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14. Huang, H. Fang, X.; **García-Rojas, J.**; Tu, Z.; Liu, J.; Liu, X. -W.
[*A Morpho-kinematic Study of Galactic High-ADF PNe Based on the VLT/UVES Deep Spectroscopy*](#)
2026, ApJS, **282**, 63
15. Wesson, R.; Drew, J. E.; Barlow, M. J.; **García-Rojas, J.**; et al. (+22 authors)

16. Tu, Z; Fang, X.; **García-Rojas, J.**; Williams, R.; Liu, J.
[*PyEMILL: A New Generation Computer-aided Spectral Line Identifier II. Application on A Sample of Emission-line Objects*](#)
2026, ApJS, **282**, 32
17. Bouvis, K.; Akas, S.; Monteiro, H.; Konstantinou, L.; Boumis, P.; **García-Rojas, J.**; Gonçalves, D. R.; Aleman, I.; Monreal-Ibero, A.; Cami, J.
[*Nickel and Iron-Rich Clumps in Planetary Nebulae: New Discoveries and Emission Line Diagnostics*](#)
2025, A&A, **700**, A155
18. Bouvis, K.; Akas, S.; Monteiro, H.; Konstantinou, L.; Boumis, P.; **García-Rojas, J.**; Gonçalves, D. R.; Monreal-Ibero, A.; Aleman, I.; Gourgouliatos, K. N.
[*Dissecting NGC 3132: Tracing the mass-loss history of the southern ring planetary nebula*](#)
2025, MNRAS, **541**, 1628
19. Konstantinou, L.; Akas, S.; **García-Rojas, J.**; et al. (+7 authors)
[*Integral field spectroscopy of the planetary nebula NGC 3242 and the puzzling nature of its low ionization structures*](#)
2025, A&A, **697**, A227
20. Orte-García, M.; Esteban, C.; Méndez-Delgado, J. E.; **García-Rojas, J.**; Arellano-Córdova, K.; Rosales-Ortega, F. F.; Reyes-Rodríguez, E.
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2025, A&A, **697**, A130
21. Esteban, C.; Méndez-Delgado, J. E.; **García-Rojas, J.** et al. (+6 authors)
[*Alpha-element abundance patterns in star-forming regions of the local Universe*](#)
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22. Méndez-Delgado, J. E.; Skillman, E. D.; Aver, E.; Morisset, C.; Esteban, C.; **García-Rojas, J.**; et al. (+8 authors)
[*Generalized Te\(\[O III\]\)–Te\(He I\) Discrepancies in Ionized Nebulae: Possible Evidence of Case B Deviations and Temperature Inhomogeneities*](#)
2025, ApJ, **986**, 74, id. 74
23. Méndez-Delgado, J. E.; Kreckel, K.; Esteban, C.; **García-Rojas, J.**; et al. (+9 authors)
[*Gas-phase Fe/O and Fe/N abundances in Star-Forming Regions. Relations between nucleosynthesis, metallicity and dust*](#)
2024, A&A, **690**, A248
24. Gómez-Llanos, V.; **García-Rojas, J.**; Morisset, C.; Monteiro, H.; Jones, D.; Wesson, R.; Boffin, H. M. J.; Corradi, R. L. M.
[*MUSE spectroscopy of the high abundance discrepancy planetary nebula NGC 6153*](#)
2024, A&A, **689**, A228
25. Akas, S.; Monteiro, H.; Walsh, J. R.; Konstantinou, L.; Gonçalves, D. R.; **García-Rojas, J.**; Boumis, P.; Aleman, I.
[*Detection of the \[C II\] 8727 emission line. The case of low-ionization structures in NGC 7009*](#)
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26. Reyes-Rodríguez, E.; Méndez-Delgado, J. E.; **García-Rojas, J.**; Binette, L.; Nemer, A.; Esteban, C.; Kreckel, K.
[*Excitation mechanisms of C II optical permitted lines in ionized nebulae*](#)
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2024, Nature Astronomy, **8**, 275
28. Méndez-Delgado, J. E.; Esteban, C.; **García-Rojas, J.**; et al. (+6 authors)
[*Density biases and temperature relations for DESIRED H II regions*](#)
2023, MNRAS, **523**, 2952
29. Agudo, I. et al. (+103 authors, including **García-Rojas, J.**)
[*Panning for gold, but finding helium: Discovery of the ultra-stripped supernova SN 2019wxt from gravitational-wave follow-up observations*](#)
2023, A&A, **675**, A201
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[*Temperature inhomogeneities cause the abundance discrepancy in H II regions*](#)

2023, Nature, **618**, 249

31. Morisset, C.; Corradi, R. L. M.; **García-Rojas, J.**; Mampaso, A.; Jones, D.; Kwitter, K. B.; Magrini, L.; Villaver, E.
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[*Atomic Data Assessment with PyNeb: Radiative and Electron Impact Excitation Rates for \[Fe II\] and \[Fe III\]*](#)
2023, Atoms, **11**, 63
33. De Marco, O. et al. (+68 authors, including **García-Rojas, J.**)
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34. Domínguez-Guzmán, G.; Rodríguez, M.; **García-Rojas, J.**; Esteban, C.; Toribio San Cipriano, L.
[*The homogeneity of chemical abundances in H II regions of the Magellanic Clouds*](#)
2022, MNRAS, **517**, 4497
35. Jones, D.; Kabath, P.; **García-Rojas, J.**; Hanus, J.; Jakubik, M.; Janik, J.; Nagy, R.; Toth, J.
[*Per aspera ad astra simul: Through difficulties to the stars together*](#)
2022, Astronomy Education Journal, **2**, 3
36. Méndez-Delgado, J. E.; Esteban, C.; **García-Rojas, J.**; Henney, W. J.
[*Photoionized Herbig-Haro objects in the Orion Nebula through deep high-spectral resolution spectroscopy - III. HH 514*](#)
2022, MNRAS, **514**, 744
37. Esteban, C.; Méndez-Delgado, J. E.; **García-Rojas, J.**; Arellano-Córdova, K. Z.
[*About Metallicity Variations in the Local Galactic Interstellar Medium*](#)
2022, ApJ, **931**, 92
38. Akas, S.; Monteiro, H.; Walsh, J. R.; **García-Rojas, J.**; et al. (+10 authors)
[*Spectroscopic analysis tool for intEgral fieLd unIt daTacubEs \(SATELLITE\): case studies of NGC 7009 and NGC 6778 with MUSE*](#)
2022, MNRAS, **512**, 2202
39. **García-Rojas, J.**; Morisset, C.; Jones, D.; Wesson, R.; Boffin, H. M. J.; Monteiro, H.; Corradi, R. L. M.; Rodríguez-Gil, P.
[*MUSE spectroscopy of planetary nebulae with high abundance discrepancies*](#)
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40. Méndez-Delgado, J. E.; Amayo, A.; Arellano-Córdova, K. Z.; Esteban, C.; **García-Rojas, J.**; Carigi, L.; Delgado-Inglada, G.
[*Gradients of chemical abundances in the Milky Way from H II regions: distances derived from Gaia EDR3 parallaxes and temperature inhomogeneities*](#)
2022, MNRAS, **510**, 4436
41. Jones, D.; Munday, J.; Corradi, R. L. M.; Rodríguez-Gil, P.; et al. (+9 authors, including **García-Rojas, J.**)
[*The post-common-envelope binary central star of the planetary nebula Ou 5: a doubly eclipsing post-red-giant-branch system*](#)
2022, MNRAS, **510**, 3102
42. Galera-Rosillo, R.; Mampaso, A.; Corradi, R. L. M.; **García-Rojas, J.**; Balick, B.; Jones, D.; Kwitter, K. B.; Magrini, L.; Villaver, E.
[*On the most luminous planetary nebulae of M 31*](#)
2022, A&A, **657**, A71
43. Méndez-Delgado, J. E.; Henney, W. J.; Esteban, C.; **García-Rojas, J.**; Mesa-Delgado, A.; Arellano-Córdova, K. Z.
[*Photoionized Herbig-Haro Objects in the Orion Nebula through Deep High Spectral Resolution Spectroscopy. II. HH 204*](#)
2021, ApJ, **918**, id.27
44. Arellano-Córdova, K. Z.; Esteban, C.; **García-Rojas, J.**; Méndez-Delgado, J. E.
[*On the radial abundance gradients of nitrogen and oxygen in the inner Galactic disc*](#)
2021, MNRAS, **502**, 225
45. Méndez-Delgado, J. E.; Esteban, C.; **García-Rojas, J.**; Henney, W.; Mesa-Delgado, A.; Arellano-Córdova, K.
[*Photoionized Herbig-Haro objects in the Orion Nebula through deep high-spectral resolution spectroscopy I: HH 529 II and III*](#)

46. Armas-Padilla, M.; Muñoz-Darias, T.; Jiménez-Ibarra, F.; Fernández-Ontiveros, J. A.; (+5 authors, including **García-Rojas, J.**)
[*Optical spectroscopy of 4U 1812-12: an ultra-compact X-ray binary seen through an HII region*](#)
2020, A&A, **644**, A63
47. Jones, D.; Boffin, H. M. J.; Hibbert, J.; Steinmetz, T.; et al. (+7 authors, including **García-Rojas, J.**)
[*The post-common-envelope binary central star of the planetary nebula PN G283.7-05.1: A possible post-red-giant-branch planetary nebula central star*](#)
2020, A&A, **642**, A108
48. Morisset, C.; Luridiana, V.; **García-Rojas, J.**; Gómez-Llanos, V.; Bautista, M. A.; Mendoza, C.
[*Atomic Data Assessment with PyNeb*](#)
2020, Atoms, **8**, 66
49. Munday, J.; Jones, D.; **García-Rojas, J.** et al. (+7 authors)
[*The post-common-envelope binary central star of the planetary nebula ETHOS 1*](#)
2020, MNRAS, **498**, 6005
50. Delgado-Inglada, G.; **García-Rojas, J.**; Stasinska, G.; Rechy-García, J. S.
[*A study of extragalactic planetary nebulae populations based on spectroscopy. I. Data compilation and first findings*](#)
2020, MNRAS, **498**, 5367
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[*The impact of strong recombination on temperature determination in planetary nebulae*](#)
2020, MNRAS Letters, **498**, L82
52. Méndez-Delgado, J. E.; Esteban, C.; **García-Rojas, J.**; Arellano-Córdova, K. Z.; Valerdi, M.
[*Helium abundances and its radial gradient from the spectra of H II regions and ring nebulae of the Milky Way*](#)
2020, MNRAS, **496**, 2726
53. Arellano-Córdova, K. Z.; Esteban, C.; **García-Rojas, J.**; Méndez-Delgado, J. E.
[*The Galactic radial abundance gradients of C, N, O, Ne, S, Cl, and Ar from deep spectra of H II regions*](#)
2020, MNRAS, **496**, 1051
54. Amayo, A.; Delgado-Inglada, G.; **García-Rojas, J.**
[*Ionization correction factors for sodium, potassium, and calcium in planetary nebulae*](#)
2020, MNRAS, **492**, 950
55. Esteban, C.; Bresolin, F.; **García-Rojas, J.**; Toribio San Cipriano, L.
[*Carbon, nitrogen, and oxygen abundance gradients in M101 and M31*](#)
2020, MNRAS, **491**, 2137
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[*First Evidence of Enhanced Recombination in Astrophysical Environments and the Implications for Plasma Diagnostics*](#)
2019, ApJL, **887**, L9
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[*Accretion and outflow in V404 Cyg*](#)
2019, MNRAS, **488**, 1356
58. Wesson, R.; Jones, D.; **García-Rojas, J.**; Boffin, H. M. J.; Corradi, R. L. M.
[*Close binaries and the abundance discrepancy problem in planetary nebulae*](#)
2018, Galaxies, **6**, 110
Refereed proceeding
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[*Chemistry in the dIrr galaxy Leo A*](#)
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[*Confirmation of the link between central star binarity and extreme abundance discrepancy factor in planetary nebulae*](#)
2018, MNRAS, **480**, 4589
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[*Revisiting the radial abundance gradients of nitrogen and oxygen in the Milky Way*](#)

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[*Neutron-Capture elements in planetary nebulae: first detections of near-Infrared \[Te III\] and \[Br V\] emission lines*](#)
2018, ApJ, **861**, L8
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[*C/O ratios in planetary nebulae with dual-dust chemistry from faint optical recombination lines*](#)
2018, MNRAS, **473**, 4476
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[*The kinematic behaviour of optical recombination lines and collisionally excited lines in Galactic planetary nebulae*](#)
2017, MNRAS, **472**, 1182
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2017, MNRAS, **471**, 3529
66. Madonna, S.; **García-Rojas, J.**; Sterling, N. C.; Delgado-Inglada, G.; Mesa-Delgado, A.; Luridiana, V.; Roederer, I. U.; Mashburn, A. L.
[*Neutron-capture element abundances in the planetary nebula NGC 5315 from deep optical and near-infrared spectrophotometry*](#)
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[*The radial abundance gradient of oxygen towards the Galactic anti-centre*](#)
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68. Toribio San Cipriano, L.; Domínguez-Guzmán, G.; Esteban, C.; **García-Rojas, J.**; Mesa-Delgado, A.; Bresolin, F.; Rodríguez, M.; Simón-Díaz, S.
[*Carbon and oxygen in H II regions of the Magellanic Clouds: abundance discrepancy and chemical evolution*](#)
2017, MNRAS, **467**, 3759
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[*Identification of Near-infrared \[Se III\] and \[Kr VI\] Emission Lines in Planetary Nebulae*](#)
2017, ApJ, **840**, 80
70. **García-Rojas, J.**; Corradi, R. L. M.; Monteiro, H.; Jones, D.; Rodríguez-Gil, P.; Cabrera-Lavers, A.
[*Imaging the elusive H-poor gas in the high ADF planetary nebulae NGC 6778*](#)
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73. Delgado-Inglada, G.; Mesa-Delgado, A.; **García-Rojas, J.**; Rodríguez, M.; Esteban, C.
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[*The radial abundance gradient of chlorine in the Miky Way*](#)
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[*Detection of a large Be circumstellar disk during X-ray quiescence of XTE J1946+274*](#)
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[*Binarity and the abundance discrepancy problema in planetary nebulae*](#)
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2014, Rev. Mex. Astron. & Astrofis., **50**, 329
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[*Carbon and oxygen abundances from recombination lines in low-metallicity star-forming galaxies. Implications for chemical evolution*](#)
2014, MNRAS, **443**, 624
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[*The planetary nebula IPHASXJ211420.0+434136 \(Ou5\): insights into common-envelope dynamical and chemical evolution*](#)
2014, MNRAS, **441**, 2799
84. Camero, A.; Zurita, C.; Gutierrez Soto, J.; Ozbey Arabaci, M.; et al. (+5 authors, including **García-Rojas, J.**)
[*Recent activity of the Be/X-ray binary system SAX J2103.5+4545*](#)
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2014, A&A, **568**, A82
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[*The Trace of the CNO Cycle in the Ring Nebula NGC 6888*](#)
2014, ApJ, **785**, 100
87. Esteban, C.; **García-Rojas, J.**; Mesa-Delgado, A.; Toribio San Cipriano, L.
[*Deep high spectral resolution spectroscopy and chemical composition of ionized nebulae*](#)
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<https://adriandelorenzocaceres.wordpress.com/gaveta-de-astrofisica/>

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